

SCIBENCH ANONYMIZED REPLICATION DOSSIER

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Anonymous replication dossier: transformed regression

Scientific objective

Quantify the bias introduced when noisy exponential-decay observations are log-transformed before regression. Compare fitting in the original observation space with ordinary and weighted least squares in transformed space.

Model and estimators

Use the first-order decay model

[EQUATION]

$$y(t) = y_0 \exp(-k t).$$

The transformed relation is

[EQUATION]

$$\log(y(t)) = -k t + \log(y_0).$$

For every simulated data set, estimate k in two ways:

1. Fit $y(t)$ directly to the exponential model by nonlinear least squares, treating both k and y_0 as free parameters.
2. Fit $\log(y(t))$ against t by linear least squares and negate the fitted slope.

The unweighted comparison assumes constant additive uncertainty in the original observation space. For the weighted comparison, propagate a common observation standard deviation through the logarithm. The transformed uncertainty for an observation y_i is $\sigma_{\log,i} = \sigma_y / y_i$, so the corresponding WLS weight used by the reference experiment is proportional to $1 / \sigma_{\log,i}$. This is the historical weighting convention evaluated by this replication task. Use the same generated observations for the paired nonlinear and transformed fits.

Monte Carlo protocol

Set $k = 0.15 \text{ s}^{-1}$ and $y_0 = 7.5 \text{ mol m}^{-3}$. Generate 2^{15} independent noisy data sets using NumPy's default random generator initialized with seed 1, at $t = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ seconds. Use additive Gaussian measurement noise with standard deviation 0.3 mol m^{-3} for the uncertainty-aware comparison. Reject or otherwise handle non-positive noisy concentrations consistently before taking a logarithm; do not silently replace the requested number of successful replicates.

For every fitted rate constant, form the dimensionless normalized estimate $r = \hat{k} / k$, where the denominator is the true rate constant $k = 0.15 \text{ s}^{-1}$. Compute the requested means and the 2.5th and 97.5th percentiles from the empirical distribution of r , not from the raw fitted $\hat{k}$ values. The six artifacts correspond to the transformed and direct-fit normalized summaries for the OLS and WLS comparisons. Write each requested artifact separately.

[MASKED: all reference estimates and interval endpoints are withheld.]